GREAT LAKES CHEMICAL PROCESSING (GLCP)

FINAL INSPECTION REPORT

Ultrasonic Thickness Survey
Hydrotreater Unit 200 — Pipe Racks and Pipe Bridge

Report Number: UT-26-0217-200
Revision: 0
Report Date: 2026-03-05

Client: Great Lakes Chemical Processing (GLCP)
Site: GLCP — North Plant, Mississauga, Ontario
Job / Work Order: GLCP-UT-2026-0217

Survey Dates: 2026-02-17 to 2026-02-21
Method / Code Basis: UT Thickness (ASME V Article 5)
Scope: 10 CMLs across 2 piping lines

Prepared By: UMA AI Inspection Intelligence (Sample Dataset)
Contractor: UMA NDT Services Ltd.
Client Contact: A. Chen, Inspection Supervisor

A. Report Front Matter

1) Executive Summary

This report documents an ultrasonic thickness (UT) survey completed on selected piping circuits in Hydrotreater Unit 200. A total of 10 corrosion monitoring locations (CMLs) across 2 lines were inspected between 2026-02-17 and 2026-02-21. Thickness readings were compared against nominal wall thickness and minimum required thickness (t-min) values to support integrity trending and inspection planning in accordance with the referenced codes and GLCP procedure.

Key outcomes: **1 CML was classified as CRITICAL** (measured below t-min) and required immediate notification and disposition. **2 CMLs were classified as Alert** (measured below 110% of t-min). **1 CML was classified as Monitor** due to elevated short-term corrosion rate (> 10 mpy). All other CMLs were classified as Acceptable.

2) Scope of Work

Unit / Area: Hydrotreater Unit 200 (Pipe Racks and Pipe Bridge)

Lines / Circuits: 200-L-1102, 200-L-1148

CML Count: 10 (5 per line)

Survey Dates: 2026-02-17 to 2026-02-21

Method: UT Thickness (ASME V Article 5)

Data Requirements: Minimum of 4 readings per CML at approximately 90° intervals; minimum value recorded per GLCP-NDT-001 Rev 3.

3) Reference Documents

- API 570 — Piping Inspection Code, in-service inspection, rating, repair, and alteration.
- API 574 — Inspection Practices for Piping System Components.
- ASME B31.3 — Process Piping (t-min basis).
- ASME Boiler & Pressure Vessel Code Section V, Article 5 — Ultrasonic Examination Methods.
- GLCP-NDT-001 Rev 3 — UT Thickness Survey Procedure (company requirement).
- API 579-1/ASME FFS-1 — Fitness-For-Service (when disposition requires evaluation).

B. Personnel & Qualifications

5–8) Personnel, Certification, and Safety

Name	Role	Cert Level	Cert No.	Employer	Safety Orientation
Jordan Patel	Lead UT Inspector	UT Level II	UT-LL2-18427	UMA NDT Services Ltd.	Completed (GLCP Site)
Sofia Martinez	UT Inspector	UT Level II	UT-LL2-19311	UMA NDT Services Ltd.	Completed (GLCP Site)
Ethan Okafor	Assistant	Trainee (under supervision)	UT-TR-0774	UMA NDT Services Ltd.	Completed (GLCP Site)

Certification records and training evidence are included in Appendix H-5. Minimum qualification requirements (UT Level II with documented experience) were met for personnel performing measurements; trainees worked under direct supervision.

C. Equipment & Calibration

9–14) Equipment Identification, Certificates, and Settings

Item	Model / Description	Serial / ID	Calibration Certificate
UT gauge	Olympus 38DL PLUS	38DL-19-004821	CAL-38DL-2025-11-033 (valid to 2026-11-30)
Transducer	Dual element 5 MHz, 0.375 in	DE-5M-0375-2219	Included in gauge kit cert
Cal block	IIW Type 1 (steel)	IIW-STEEL-1142	BLK-1142-2025-10-018 (traceable to NIST)

Material velocity settings used:

Material	Velocity Setting (in/µs)
Carbon steel (CS)	0.2330
Stainless steel (SS 316)	0.2280

Daily calibration verification record (tolerance ±0.002 in):

Date	Time	Operator	Block Thk (in)	Reading 1	Reading 2	Reading 3	Within Tolerance
2026-02-17	07:30	JP	0.500	0.500	0.500	0.499	Yes
2026-02-18	07:30	SM	0.500	0.500	0.500	0.499	Yes
2026-02-19	07:30	JP	0.500	0.499	0.501	0.501	Yes
2026-02-20	07:30	SM	0.500	0.501	0.499	0.501	Yes
2026-02-21	07:30	JP	0.500	0.501	0.501	0.500	Yes

D. Inspection Results — Data Tables

15–22) UT thickness results are tabulated by line and CML. For each CML, a minimum of four (4) readings were collected at approximately 90° intervals. The minimum reading is recorded as the controlling thickness for trending and acceptance.

Line / Circuit: 200-L-1102 — H2 recycle | NPS 6" | CS Sch 80

CML	Location	Comp.	Nom. (in)	t-min (in)	Prev 2019	Current Min (in)	Category
CML-200-L-1102-01	At low point drain	Straight pipe	0.237	0.154	0.181	**0.146**	**CRITICAL**
CML-200-L-1102-02	At low point drain	Tee	0.432	0.281	0.346	0.325	Acceptable
CML-200-L-1102-03	Near PSV 1102A	Flange	0.337	0.219	0.262	0.248	Acceptable
CML-200-L-1102-04	N. rack @ Bay 3	Straight pipe	0.432	0.281	0.361	0.337	Acceptable
CML-200-L-1102-05	At low point drain	Reducer	0.432	0.281	0.388	0.366	Acceptable

CML	R1 (in)	R2 (in)	R3 (in)	R4 (in)	Date	Time	Op.	Gauge S/N
CML-200-L-1102-01	0.146	0.152	0.156	0.158	2026-02-17	09:15	JP	38DL-19-004821
CML-200-L-1102-02	0.325	0.330	0.334	0.341	2026-02-17	11:15	SM	38DL-19-004821
CML-200-L-1102-03	0.264	0.266	0.267	0.248	2026-02-17	13:15	JP	38DL-19-004821
CML-200-L-1102-04	0.356	0.337	0.344	0.344	2026-02-18	09:15	SM	38DL-19-004821
CML-200-L-1102-05	0.366	0.380	0.375	0.383	2026-02-18	11:15	JP	38DL-19-004821

Line / Circuit: 200-L-1148 — Sour water | NPS 4" | CS Sch 40

CML	Location	Comp.	Nom. (in)	t-min (in)	Prev 2019	Current Min (in)	Category
CML-200-L-1148-01	Downstream of exchanger E-201	Flange	0.337	0.219	0.290	0.268	Acceptable
CML-200-L-1148-02	Near PSV 1102A	Elbow 90°	0.237	0.154	0.292	0.172	Monitor
CML-200-L-1148-03	N. rack @ Bay 3	Tee	0.237	0.154	0.174	**0.155**	**Alert**
CML-200-L-1148-04	Near PSV 1102A	Elbow 90°	0.188	0.122	0.141	**0.129**	**Alert**
CML-200-L-1148-05	Near PSV 1102A	Flange	0.337	0.219	0.306	0.299	Acceptable

CML	R1 (in)	R2 (in)	R3 (in)	R4 (in)	Date	Time	Op.	Gauge S/N
CML-200-L-1148-01	0.268	0.271	0.272	0.268	2026-02-18	13:15	SM	38DL-19-004821
CML-200-L-1148-02	0.172	0.176	0.180	0.182	2026-02-19	09:15	JP	38DL-19-004821
CML-200-L-1148-03	0.156	0.155	0.164	0.161	2026-02-19	11:15	SM	38DL-19-004821
CML-200-L-1148-04	0.142	0.140	0.145	0.129	2026-02-19	13:15	JP	38DL-19-004821
CML-200-L-1148-05	0.302	0.299	0.313	0.308	2026-02-20	09:15	SM	38DL-19-004821

E. Trending & Corrosion Analysis

23–27) Corrosion Rates, Remaining Life, and Next Inspection

Corrosion rates were calculated using historical thickness data (2015 and 2019 surveys) and the current survey (2026). Rates are reported in mils per year (mpy), where 1 mil = 0.001 in.

Definitions (API 570 §7.1 basis):

- Short-term corrosion rate (mpy) = $(T_{2019} - T_{2026}) \times 1000 / \Delta t_{2019–2026}$
- Long-term corrosion rate (mpy) = $(T_{2015} - T_{2026}) \times 1000 / \Delta t_{2015–2026}$
- Remaining life (yr) = $(T_{2026} - \text{t-min}) / (CR_{used} / 1000)$

Recommended next inspection date (API 570 §6): interval set to one-half of remaining life, bounded to a minimum of 6 months and maximum of 5 years.

Line	CML	t-min (in)	Current (in)	Prev 2019	CR Short (mpy)	CR Long (mpy)	Rem. Life (yr)	Next Inspection
200-L-1102	CML-200-L-1102-01	0.154	**0.146**	0.181	5.00	4.54	**0.0**	**2026-08-22**
200-L-1102	CML-200-L-1102-02	0.281	0.325	0.346	3.00	3.09	14.2	2031-02-21
200-L-1102	CML-200-L-1102-03	0.219	0.248	0.262	2.00	3.90	7.4	2029-11-03
200-L-1102	CML-200-L-1102-04	0.281	0.337	0.361	3.43	3.54	15.8	2031-02-21
200-L-1102	CML-200-L-1102-05	0.281	0.366	0.388	3.14	5.81	14.6	2031-02-21
200-L-1148	CML-200-L-1148-01	0.219	0.268	0.290	3.14	5.45	9.0	2030-08-22
200-L-1148	CML-200-L-1148-02	0.154	0.172	0.292	17.13	16.34	1.1	2026-09-09
200-L-1148	CML-200-L-1148-03	0.154	**0.155**	0.174	2.71	5.72	**0.2**	**2026-08-22**
200-L-1148	CML-200-L-1148-04	0.122	**0.129**	0.141	1.71	3.45	**2.0**	**2027-02-21**
200-L-1148	CML-200-L-1148-05	0.219	0.299	0.306	1.00	2.09	38.3	2031-02-21

Note: Long-term corrosion rates were computed using 2015 historical thickness values; raw historical inputs are retained in the dataset.

F. Findings & Dispositions

28–34) Categorized Findings and Required Actions

Findings are categorized per GLCP-NDT-001 Rev 3 criteria and code expectations. Supporting evidence (photos, notifications, and notes) is referenced in the Appendices.

Category	CML	Line	Current Min (in)	t-min (in)	Basis	Disposition / Notes
CRITICAL	CML-200-L-1102-01	200-L-1102	**0.146**	0.154	Below t-min	Isolate and repair/replace section; perform immediate engineering assessment. API 579 FFS evaluation recommended.
Alert	CML-200-L-1148-03	200-L-1148	**0.155**	0.154	Below 110% t-min	Plan repair or increase monitoring at next outage; validate thickness with follow-up UT.
Alert	CML-200-L-1148-04	200-L-1148	**0.129**	0.122	Below 110% t-min	Plan repair or increase monitoring at next outage; validate thickness with follow-up UT.
Monitor	CML-200-L-1148-02	200-L-1148	0.172	0.154	CR > 10 mpy	Investigate corrosion mechanism; consider process review and add targeted CMLs. Engineering review if trend continues.
Acceptable	CML-200-L-1102-02	200-L-1102	0.325	0.281	Meets criteria	No action required beyond routine inspection interval.
Acceptable	CML-200-L-1102-03	200-L-1102	0.248	0.219	Meets criteria	No action required beyond routine inspection interval.
Acceptable	CML-200-L-1102-04	200-L-1102	0.337	0.281	Meets criteria	No action required beyond routine inspection interval.
Acceptable	CML-200-L-1102-05	200-L-1102	0.366	0.281	Meets criteria	No action required beyond routine inspection interval.
Acceptable	CML-200-L-1148-01	200-L-1148	0.268	0.219	Meets criteria	No action required beyond routine inspection interval.
Acceptable	CML-200-L-1148-05	200-L-1148	0.299	0.219	Meets criteria	No action required beyond routine inspection interval.

G. Recommendations

35–38) Recommended Actions and Intervals

Recommendations are based on measured thickness, calculated corrosion rates, and observed field conditions. Engineering remains responsible for final determination of inspection intervals and dispositions.

Line	CML	Cat.	CR Short (mpy)	Rem. Life (yr)	Rec. Interval	Next Date	Recommendation / Referral
200-L-1102	CML-200-L-1102-01	**CRITICAL**	5.00	**0.0**	0.5 yr	**2026-08-22**	Isolate and repair/replace section; perform immediate engineering assessment. API 579 FFS evaluation recommended.
200-L-1102	CML-200-L-1102-02	Acceptable	3.00	14.2	5.0 yr	2031-02-21	Maintain routine interval; no additional actions.
200-L-1102	CML-200-L-1102-03	Acceptable	2.00	7.4	3.7 yr	2029-11-03	Maintain routine interval; no additional actions.
200-L-1102	CML-200-L-1102-04	Acceptable	3.43	15.8	5.0 yr	2031-02-21	Maintain routine interval; no additional actions.
200-L-1102	CML-200-L-1102-05	Acceptable	3.14	14.6	5.0 yr	2031-02-21	Maintain routine interval; no additional actions.
200-L-1148	CML-200-L-1148-01	Acceptable	3.14	9.0	4.5 yr	2030-08-22	Maintain routine interval; no additional actions.
200-L-1148	CML-200-L-1148-02	Monitor	17.13	1.1	0.6 yr	2026-09-09	Investigate corrosion mechanism; consider process review and add targeted CMLs. Engineering review if trend continues.
200-L-1148	CML-200-L-1148-03	**Alert**	2.71	**0.2**	0.5 yr	**2026-08-22**	Plan repair or increase monitoring at next outage; validate thickness with follow-up UT.
200-L-1148	CML-200-L-1148-04	**Alert**	1.71	**2.0**	1.0 yr	**2027-02-21**	Plan repair or increase monitoring at next outage; validate thickness with follow-up UT.
200-L-1148	CML-200-L-1148-05	Acceptable	1.00	38.3	5.0 yr	2031-02-21	Maintain routine interval; no additional actions.

H. Appendices

39–47) Supporting Records and Photo Log

Appendices below provide supporting evidence and records referenced throughout the report. For this sample report, photographs and certificates are represented as placeholders.

Appendix	Description	Status
H-1	Raw UT data export (gauge CSV)	Included (placeholder)
H-2	Photographs — CML location (each CML)	Included (placeholder)
H-3	Photographs — gauge display (CRITICAL and Alert)	Included (placeholder)
H-4	Isometric drawings with CML locations marked	Included (placeholder)
H-5	Personnel certification records (UT)	Included (placeholder)
H-6	Equipment calibration certificates (traceable to NIST)	Included (placeholder)
H-7	Photographs — surface condition / anomalies	Included (placeholder)
H-8	Photographs — inaccessible locations (if any)	Included (placeholder)
H-9	Photographs — daily calibration setup	Included (placeholder)

Photo log (sample):

Photo ID	CML	Photo Type	Description / Notes
P-001	CML-200-L-1102-01	CML Location	Wide shot and close-up of At low point drain (tagging visible).
P-002	CML-200-L-1102-01	Gauge Display	Gauge screen showing min thickness 0.146 in.
P-003	CML-200-L-1102-02	CML Location	Wide shot and close-up of At low point drain (tagging visible).
P-004	CML-200-L-1102-03	CML Location	Wide shot and close-up of Near PSV 1102A (tagging visible).
P-005	CML-200-L-1102-04	CML Location	Wide shot and close-up of N. rack @ Bay 3 (tagging visible).
P-006	CML-200-L-1102-05	CML Location	Wide shot and close-up of At low point drain (tagging visible).
P-007	CML-200-L-1148-01	CML Location	Wide shot and close-up of Downstream of exchanger E-201.
P-008	CML-200-L-1148-02	CML Location	Wide shot and close-up of Near PSV 1102A (tagging visible).
P-009	CML-200-L-1148-03	CML Location	Wide shot and close-up of N. rack @ Bay 3 (tagging visible).
P-010	CML-200-L-1148-03	Gauge Display	Gauge screen showing min thickness 0.155 in.
P-011	CML-200-L-1148-04	CML Location	Wide shot and close-up of Near PSV 1102A (tagging visible).
P-012	CML-200-L-1148-04	Gauge Display	Gauge screen showing min thickness 0.129 in.
P-013	CML-200-L-1148-05	CML Location	Wide shot and close-up of Near PSV 1102A (tagging visible).

Sample photo placeholders (for illustration):

Placeholder: CML Location Photo

[Insert field photo here]

Placeholder: Gauge Display Photo

[Insert gauge screen photo here]

Placeholder: Calibration Setup Photo

[Insert daily calibration photo here]

Placeholder: Marked Isometric Excerpt

[Insert marked-up isometric here]